

EXPERIMENTAL RESULTS

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

1. Dataset Summary

The final experiment was conducted using the project dataset customer_churn_sample_5000.csv.

Property	Observed value
Total records	5,000
Total columns	13
Churned customers	1,365 (27.3%)
Retained customers	3,635 (72.7%)
Missing values before cleaning	150
Duplicate rows	0
Missing target labels	0
Input features used	10
Encoded model features	22

All 5,000 records were retained after the cleaning stage. Missing feature values were handled during preprocessing, and customer identifiers and configured non-predictive fields were excluded from modelling.

2. Experimental Setup

The dataset was divided using an 80/20 stratified train-test split, producing 4,000 training observations and 1,000 held-out test observations.

Class-weighted learning was used to address the class imbalance caused by the 27.3% churn class. The same preprocessing and evaluation workflow was applied to all six classification models.

The models evaluated were: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and CatBoost.

Five-fold cross-validation using ROC-AUC was used for model selection on the training partition. Final performance was then measured on the held-out test set.

3. Model Performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	CV ROC-AUC	CV Std.	Time (s)
Logistic Regression	0.698	0.4611	0.6300	0.5325	0.7616	0.5447	0.7706	0.0163	0.53
Decision Tree	0.671	0.4330	0.6630	0.5239	0.7180	0.4786	0.7079	0.0101	0.65
Random Forest	0.744	0.5521	0.3297	0.4128	0.7278	0.5250	0.7451	0.0244	6.16
Gradient Boosting	0.714	0.4820	0.6347	0.5489	0.7635	0.5546	0.7533	0.0199	9.06
XGBoost	0.711	0.4770	0.6081	0.5346	0.7443	0.5410	0.7447	0.0207	7.41
CatBoost	0.705	0.4676	0.5824	0.5188	0.7484	0.5417	0.7505	0.0190	11.21

Model Selection

The application selected Logistic Regression using five-fold cross-validation ROC-AUC, with a CV ROC-AUC of 0.7706.

On the held-out test set, Gradient Boosting produced the highest ROC-AUC at 0.7635 and the highest PR-AUC at 0.5546. Random Forest produced the highest accuracy at 0.744, while Decision Tree produced the highest recall at 0.6630.

These results demonstrate why churn prediction should be evaluated using multiple complementary metrics rather than accuracy alone.

4. Churn Risk Prediction

The final batch prediction run classified 2,148 of 5,000 customers as predicted churners.

Risk level	Customers
Low	1,642
Medium	1,941
High	889
Very High	528
Total	5,000

The application converts predicted churn probabilities into risk categories so that analysts can examine customers according to different levels of predicted risk.

The risk thresholds used by the application are implementation defaults and should be validated against actual retention costs before being used for operational decisions.

5. Feature Importance and Explainability

The feature-importance output indicated that contractual and service-related variables were prominent signals in the trained model. In the displayed XGBoost feature-importance results, contract, region, payment method, internet service, and satisfaction score appeared among the highest-ranked features.

These results describe predictive associations within the trained model. They should not be interpreted as evidence that any individual feature directly causes customer churn.

The application also provides local contribution explanations for individual predictions, allowing users to inspect which input characteristics contributed to a particular churn-risk estimate.

6. Key Observations

- The dataset contains a meaningful class imbalance, with churn representing 27.3% of the records.
- Different models show different trade-offs between accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.
- Logistic Regression was selected by the cross-validation model-selection procedure.
- Gradient Boosting achieved the highest held-out ROC-AUC and PR-AUC in the reported comparison.
- Random Forest achieved the highest accuracy but had substantially lower recall than Logistic Regression, Decision Tree, and Gradient Boosting.
- The risk-level output extends the system beyond a binary churn prediction by representing different probability ranges.
- Feature-importance and local explanation outputs improve the interpretability of the prediction workflow.
- Batch prediction allows the trained workflow to score a complete customer file rather than requiring individual manual predictions.

7. Limitations of the Reported Experiment

- The reported results are based on one 5,000-row dataset and one held-out train-test split. Therefore, the results should not be treated as evidence that the same performance will generalize to other customer populations.
- The dataset's external provenance was not documented in the supplied experiment materials, so no unsupported public dataset source is claimed here.
- The risk thresholds are application defaults rather than business-optimized thresholds.

- Feature-importance and SHAP-style explanations describe model behaviour and should not be interpreted as causal relationships.
- Further validation using larger datasets, repeated or time-based splits, hyperparameter tuning, probability calibration, cost-sensitive threshold optimization, and retention-action evaluation would strengthen the study.

8. Reproducibility Note

The values in this document correspond to the final experiment reported in the project research paper. They should be updated if the model pipeline, dataset, preprocessing configuration, or final experiment is changed.

For reproducibility, the GitHub repository should ideally contain source code, requirements.txt, dataset documentation, model-training instructions, evaluation outputs, dashboard instructions, screenshots of the final application, and this results document.